

$$\text{binomial therom: } (x+a)^r = \sum_{b=0}^r \left[\binom{r}{b} x^b a^{r-b} \right]$$

let $a = 1$, 1^n always $=1$ so all x terms are just multiplied by 1

$$(x+1)^r = \sum_{b=0}^r \left[\binom{r}{b} x^b 1 \right]$$

let $x' = x - 1$ to cancel the $+1$ and produce x^r

$$((x-1)+1)^r = x^r = \sum_{b=0}^r \left[\binom{r}{b} (x-1)^b \right]$$

bring out the $(x-1)^r$ to get recursssion (by applying the identity to that term)

$$x^r = \sum_{b=0}^{r-1} \left[\binom{r}{b} (x-1)^b \right] + (x-1)^r$$

$$x^r = \sum_{b=0}^{r-1} \left[\binom{r}{b} (x-1)^b \right] + \sum_{b=0}^{r-1} \left[\binom{r}{b} (x-2)^b \right] + \sum_{b=0}^{r-1} \left[\binom{r}{b} (x-3)^b \right] + (x-3)^r$$

$$x^r = \sum_{b=0}^{r-1} \left[\binom{r}{b} (x-1)^b \right] + \sum_{b=0}^{r-1} \left[\binom{r}{b} (x-2)^b \right] + \dots + \sum_{b=0}^{r-1} \left[\binom{r}{b} (x-x)^b \right] + (x-x)^r$$

$$x^r = \sum_{b=0}^{x-1} \left[\sum_{c=0}^{r-1} \left[\binom{r}{c} b^c \right] \right] + 0^r$$

apply the identity to the b^c terms assymetrically to get sums of x^0

$$x^r = \sum_{b=0}^{x-1} \left[\sum_{c=0}^{r-1} \left[\binom{r}{c} b^c \right] \right] + 0^r$$

$$x^r = \sum_{b=0}^{x-1} \left[\sum_{c=0}^{r-1} \left[\binom{r}{0} b^0 + \binom{r}{1} b^1 + \binom{r}{2} b^2 + \dots + \binom{r}{r-1} b^{r-1} \right] \right] + 0^r$$

$$x^r = \sum_{b=0}^{x-1} \left[\binom{r}{0} b^0 + \binom{r}{1} \sum_{c=0}^{b-1} \left[\binom{1}{0} c^0 \right] + \binom{r}{2} \sum_{c=0}^{b-1} \left[\binom{2}{0} c^0 + \binom{2}{1} \sum_{d=0}^{c-1} \left[\binom{1}{0} d^0 \right] \right] + \dots + \binom{r}{r-1} b^{r-1} \right] + 0^r$$

$$x^r = \binom{r}{0} \sum_{b=0}^{x-1} [b^0] + \binom{r}{1} \binom{1}{0} \sum_{b=0}^{x-1} \sum_{c=0}^{b-1} [c^0] + \binom{r}{2} \binom{2}{0} \sum_{b=0}^{x-1} \sum_{c=0}^{b-1} [c^0] + \binom{r}{2} \binom{2}{1} \binom{1}{0} \sum_{b=0}^{x-1} \sum_{c=0}^{b-1} \sum_{d=0}^{c-1} [d^0] + \dots + \binom{r}{r-1} b^{r-1} + 0^r$$

$$x^r = \binom{r}{0} \sum_{b=0}^{x-1} [b^0] + \left(\binom{r}{1} \binom{1}{0} + \binom{r}{2} \binom{2}{0} \right) \sum_{b=0}^{x-1} \sum_{c=0}^{b-1} [c^0] + \binom{r}{2} \binom{2}{1} \binom{1}{0} \sum_{b=0}^{x-1} \sum_{c=0}^{b-1} \sum_{d=0}^{c-1} [d^0] + \dots + \binom{r}{r-1} b^{r-1} + 0^r$$

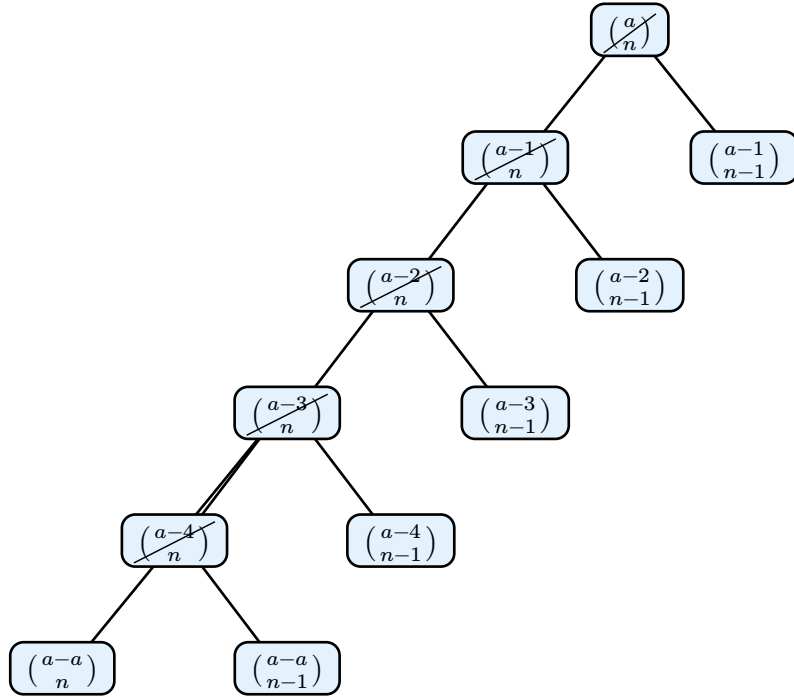
let the product and sum of these outer binom terms grouped by sum level (n) , $= D_{r,n}$

$$x^r = D_{r,1} \sum_{b=0}^{x-1} [b^0] + D_{r,2} \sum_{b=0}^{x-1} \sum_{c=0}^{b-1} [c^0] + D_{r,3} \sum_{b=0}^{x-1} \sum_{c=0}^{b-1} \sum_{d=0}^{c-1} [d^0] + \dots + D_{r,r} \overbrace{\sum_{b=0}^{x-1} \sum_{c=0}^{b-1} \sum_{d=0}^{c-1} \dots}^{r \text{ sums}} \sum_{n_n=0}^{n_{n-1}-1} [n_n^0] + 0^r$$

pascals identity

$$\binom{a}{n} = \binom{a-1}{n} + \binom{a-1}{n-1}$$

apply pascals identity recursively to lhs



$$\binom{a}{n} = \sum_{b=0}^{a-1} \left[\binom{b}{n-1} \right] + \binom{0}{n}$$

apply the identity to the inner $\binom{b}{n-1}$

$$\binom{a}{n} = \sum_{b=0}^{a-1} \left[\sum_{c=0}^{b-1} \left[\binom{c}{n-2} \right] + \binom{0}{n-1} \right] + \binom{0}{n}$$

expanding the inner term n-1 times, for n total sums, causes its inner term to be 1, as $\binom{a \geq 0}{0} = 1$

$$\binom{a}{n} = \underbrace{\sum_{b=0}^{a-1} \left[\sum_{c=0}^{b-1} \left[\sum_{d=0}^{c-1} \left[\dots \sum_{n_n=0}^{n_{n-1}-1} \left[\binom{n_n}{n-n} \right] + \binom{0}{n-(n-1)} \dots \right] \binom{0}{n-2} \right] + \binom{0}{n-1} \right]}_{n \text{ sums}} + \binom{0}{n}$$

$$\binom{a}{n} = \underbrace{\sum_{b=0}^{a-1} \left[\sum_{c=0}^{b-1} \left[\sum_{d=0}^{c-1} \left[\dots \sum_{n_n=0}^{n_{n-1}-1} \left[\binom{n_n}{0} \right] + \binom{0}{n-(n-1)} \dots \right] \binom{0}{n-2} \right] + \binom{0}{n-1} \right]}_{n \text{ sums}} + \binom{0}{n}$$

as we only expanded n-1 times the $\binom{0}{n_{n-1}}$ only reach $\binom{0}{n-(n-1)} = \binom{0}{1} = 0$, as $\binom{0}{n > 0} = 0$

if $n = 0$, the sums arent expanded and only the $\binom{0}{n}$ remains

$$\binom{a}{n} = \underbrace{\sum_{b=0}^{a-1} \left[\sum_{c=0}^{b-1} \left[\sum_{d=0}^{c-1} \left[\dots \sum_{n_n=0}^{n_{n-1}-1} [1] \right] \right] \right]}_{n \text{ sums}} + \binom{0}{n}$$

the sums in the binomial therom expansion and pascals identity expansions have the same bounds change

$$x^r = D_{r,1} \sum_{b=0}^{x-1} [1] + D_{r,2} \sum_{b=0}^{x-1} \sum_{c=0}^{b-1} [1] + D_{r,3} \sum_{b=0}^{x-1} \sum_{c=0}^{b-1} \sum_{d=0}^{c-1} [1] + \dots + D_{r,r} \overbrace{\sum_{b=0}^{x-1} \sum_{c=0}^{b-1} \sum_{d=0}^{c-1} \dots}^{r \text{ sums}} \sum_{n_n=0}^{n_{n-1}-1} [1] + 0^r$$

$$x^r = D_{r,1} \binom{x}{1} + D_{r,2} \binom{x}{2} + D_{r,3} \binom{x}{3} + \dots + D_{r,r} \overbrace{\binom{x}{r}}^{r \text{ sums}} + 0^r$$